Palaeogeography, Palaeoclimatology, Palaeoecology 490 (2018) 673–686

Contents lists available at ScienceDirect

Palaeogeography, Palaeoclimatology, Palaeoecology

journal homepage: www.elsevier.com/locate/palaeo

Palaeobiodiversity research based on stable isotopes: Correction of the sea spray effect on bone carbonate $\delta^{13}C$ and $\delta^{18}O$ by Gaussian Mixture Model clustering

Andrea Göhring[a,*], Markus Mauder[b], Marina Vohberger[a], Olaf Nehlich[c,d], Claus von Carnap-Bornheim[e], Volker Hilberg[f], Peer Kröger[b], Gisela Grupe[a]

[a] Ludwig Maximilian University, Faculty of Biology, Department of Biology I, Anthropology and Human Genomics, Großhaderner Str. 2, 82152 Planegg-Martinsried, Germany
[b] Ludwig Maximilian University, Faculty of Mathematics, Computer Science and Statistics, Department of Computer Science, Database Systems Group, Oettingenstraße 67, 80538 Munich, Germany
[c] Max Planck Institute for Evolutionary Anthropology, Human Evolution, Deutscher Platz 6, 04103 Leipzig, Germany
[d] University of British Columbia, Department of Anthropology, 6303 N. W. Marine Drive, Vancouver, BC V6Z 1T1, Canada
[e] Centre for Baltic and Scandinavian Archaeology, Gottorf Castle, Schlossinsel 1, 24837 Schleswig, Germany
[f] State Archaeological Museum, Gottorf Castle, Schlossinsel 1, 24837 Schleswig, Germany

ARTICLE INFO

Keywords:
Apatite
Collagen
GMM
Limnic influence
Sea spray

ABSTRACT

Transport of sea spray aerosol in coastal areas ("sea spray" effect) can have a marked influence on isotopic ratios of terrestrial ecosystems shifting terrestrial isotopic ratios towards unusual high values masking the original terrestrial signature. It is unclear so far if and to what extend sea spray influences other stable isotopes besides sulphur. In this study, we examined if the effect was also detectable in carbon, nitrogen, and oxygen stable isotopes of bone collagen and carbonate, respectively.

Multi-isotope data of mammals sampled from the Viking Haithabu and medieval Schleswig sites in Northern Germany were analysed according to a previously developed approximation procedure and Gaussian Mixture Model (GMM) clustering in order to quantify the sea spray effect in the isotopes under study.

While we were able to approximate an influence of the sea spray effect of at least 32.8% and 62.8% in $\delta^{13}C_{carb}$ and $\delta^{18}O_{carb}$, respectively, it was not possible to validate or approximate this effect in $\delta^{13}C_{coll}$ and $\delta^{15}N_{coll}$. Indeed, detection of the sea spray effect not only in $\delta^{34}S_{coll}$, but also in $\delta^{13}C_{carb}$ and $\delta^{18}O_{carb}$ is of particular importance for studies on both prehistoric and recent material.

GMM clustering on terrestrial herbivorous and marine piscivorous mammals was used to confirm the existing influence and to validate the approximated correction for the sea spray effect in the respective isotopic ratios ($\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$, $\delta^{34}S_{coll}$) and the correction for the limnic influence on $\delta^{15}N_{coll}$ approximated in a previous study. After correction, the clustering results markedly changed corresponding to the actual diet and habitat preference of the examined species. Although our study focused on palaeoecology, we suggest that GMM clustering also constitutes a very useful tool for modern landscape ecology based on stable isotope analyses.

1. Introduction

Palaeobiodiversity research, particularly the reconstruction of changes in the spatial distribution and abundance of vertebrate species, benefits substantially from stable isotope analyses of archaeological vertebrate bone finds. This archaeometric approach constitutes a useful tool for the assessment of important aspects of palaeobiodiversity such as feeding and habitat preference of taxa, and migration and import of vertebrate species.

Interpretation of stable isotopic signals in archaeological bones is particularly complex in brackish water environments because they offer a diversity of ecological niches characterised by a mixture of different salinity levels, from freshwater to marine, in the intertidal zone. Moreover, isotopic signatures in vertebrates that roam coastal areas can be markedly influenced by the so called "sea spray" effect. This effect is well documented for stable sulphur isotopes (e. g. Wadleigh et al., 1994), because atmospheric transport of sulphate through sea spray aerosol leads to an enrichment of ^{34}S in terrestrial systems thereby

* Corresponding author.
E-mail address: andrea.goehring@campus.lmu.de (A. Göhring).

https://doi.org/10.1016/j.palaeo.2017.11.057
Received 15 May 2017; Received in revised form 3 November 2017; Accepted 26 November 2017
Available online 28 November 2017
0031-0182/ © 2017 Elsevier B.V. All rights reserved.

mimicking marine values. As already detected by Wakshal and Nielsen (1982) sulphate is isotopically enriched close to the coast, while it becomes isotopically lighter with increasing distance from the sea. However, the sea spray effect in $\delta^{34}S$ can be detected in a coastal corridor of about 100 km (Nehlich, 2015), where individuals can be enriched in ^{34}S due to the marine sea spray signal. This also depends on the prevailing wind as demonstrated for Ireland by Zazzo et al. (2011). While the sea spray was still detectable about 100 km inland from the west coast, individuals located closer to the east coast (60–70 km) exhibited much lower average $\delta^{34}S$ values. This can be explained by the fact that the prevailing wind in the examined site (Ireland) is between south and west (Zazzo et al., 2011).

Plants capture the natural enrichment in ^{34}S in the sulphate molecules of their cells. Since only a small net fractionation against ^{34}S occurs during sulphur uptake, sulphated metabolites in plants show quite similar $\delta^{34}S$ values compared to substrate sulphur (Monaghan et al., 1999; Tcherkez and Tea, 2013). The only sources of sulphur for animals are diet and drinking water (Schmidt et al., 2015). As herbivores gain 10–40% of their sulphur supply by sulphate relatively enriched in ^{34}S and another 60 to 90% by organic sulphur relatively ^{34}S-depleted (Tanz and Schmidt, 2010), we expect a distinct influence of marine sulphate taken up by herbivores as well as in the course of the trophic chain. This is even more probable as the only reaction accompanied by fractionation of sulphur isotopes in animals is the sulphur activation for biosynthesis of sulphate esters (Schmidt et al., 2015).

We have already shown previously that it is possible to approximate the sea spray effect on collagen $\delta^{34}S$ by use of appropriate mathematical methods (Göhring et al., 2015). At the Haithabu and Schleswig sites which date from Viking to Medieval times (see below), a local sea spray influence between about 16.5 and 21.9% shifted bone $\delta^{34}S_{coll}$ of terrestrial mammals, terrestrial and freshwater birds, as well as humans towards unusual high values (Göhring et al., 2015). Uncorrected measurement data would therefore lead to biased palaeodietary and habitat reconstructions. Correction of $\delta^{34}S_{coll}$ according to the local sea spray effect leads to much more reliable dietary reconstructions, which are compatible with the actual diet or habitat of the examined taxa (Göhring et al., 2015).

The atomic weight of sulphur considerably exceeds the atomic weights of the elements carbon, nitrogen, and oxygen. It therefore appears rather unlikely that these lighter atoms would not be prone to the sea spray in a similar way. Both Richards et al. (2006) and Virginia and Delwiche (1982), for instance, indeed found some evidence for sea spray effect on $\delta^{15}N_{coll}$ values in animals and plants, respectively, manifested by an enrichment with ^{15}N. No attempts have been made so far for an approximation of the magnitude of this effect in archaeological remains. Furthermore, to our knowledge, the influence of sea spray on other isotope systems is not yet verified at all. Therefore, the aim of this study was to evaluate if there was a detectable influence of the sea spray on $\delta^{13}C_{coll}$, $\delta^{13}C_{carb}$, $\delta^{15}N_{coll}$, and $\delta^{18}O_{carb}$ values, respectively, and, if applicable, to approximate the magnitude of the effect, similar to the approximation procedure of the sea spray effect on $\delta^{34}S_{coll}$ (Göhring et al., 2015). Gaussian Mixture Model (GMM) clustering (see below) was used to detect clusters in the data set as well as to illustrate the isotopic signature of the examined specimens before and after correction.

2. Materials and methods

The large isotopic data set of animal bones recovered from the Viking Haithabu (804–1066 CE) and medieval Schleswig (1070–1350 CE) sites in Northern Germany located on the Jutland peninsula (Grupe et al., 2013; Hilberg, 2008) in close proximity to the Baltic Sea (Fig. 1) was used for an examination of the sea spray effect on stable carbon, nitrogen, and oxygen isotope values. A subsample of 111 terrestrial and 27 marine mammals was analysed (Tables S1 and S2). Another smaller subsample ($n = 33$) additionally including $\delta^{34}S$ was used for a demonstration of clustering in the five-dimensional case (Tables S1 and S2). The original isotopic data are published by Doppler et al. (2010) and Göhring et al. (2015).

Both Haithabu and Schleswig were important trading centres at their time. Consequently, a considerable amount of animals excavated at these sites was probably imported (Becker and Grupe, 2012; Doppler et al., 2010; Grupe et al., 2009). The area around Haithabu and Schleswig is particularly influenced by the brackish environment of the Baltic coast. The salinity level of the Baltic Sea increases from the northern part (Bothnian Bay) with 2–5 psu (practical salinity units) to the western part (Arkona Basin) with a salinity between 5 and 10 psu (Emeis et al., 2003; Fig. 1). As dietary reconstructions can be quite complex in the case of a brackish environment and one might also expect at least some influence of sea spray (see above), this data set is well suited for our study. However, the primarily unknown proportion of imported animals exhibiting a non-local isotopic signature has to be kept in mind.

2.1. Stable isotope analysis

The laboratory protocols for purification of bone collagen and structural carbonate including the validation of the isotopic signatures was previously published by Doppler et al. (2010). For the analysis of $\delta^{13}C_{coll}$ and $\delta^{15}N_{coll}$ 0.3–0.5 mg of the extracted gelatine weighed into tin capsules were combusted in a CHN-analyser (Carlo-Erba 1110) and afterwards analysed by mass spectrometry (Thermo Finnigan Delta Plus; Geo-Center of Northern Bavaria, University of Erlangen, Germany) (Doppler et al., 2010). Stable carbon and nitrogen isotopic ratios were expressed against the VPDB (Vienna Pee Dee Belemnite) and the AIR (atmospheric nitrogen) standard, respectively, provided by the Section of Isotope Hydrology of the International Atomic Energy Agency (IAEA) in Vienna, Austria. Replicate analysis of laboratory standards calibrated to the international standard USGS 40 ($\delta^{13}C = -26.3‰ \pm 0.10‰$, $\delta^{15}N = -4.52‰ \pm 0.08‰$, $n = 43$; IAEA) was used to test for reproducibility and accuracy of carbon and nitrogen isotope analyses.

For $\delta^{34}S_{coll}$ measurements, extracted gelatine samples were further processed at the Max-Planck-Institute for Evolutionary Anthropology in Leipzig (Germany): After ultrafiltration with a 30 kDa filter in order to remove degraded molecules, salts and other impurities (see Nehlich and Richards, 2009) samples were analysed with a mass spectrometer (Thermo Finnigan Delta V plus) coupled with an elemental analyser (Heka EuroVector) with a measurement precision of ± 0.4‰. Stable sulphur isotope values were expressed against the VCDT (Vienna Cañon-Diablo Troilite; IAEA) standard. Reproducibility of sulphur isotope analyses was monitored by replicate analyses of laboratory standards calibrated to the international standards NBS 127 ($\delta^{34}S = +20.30‰ \pm 0.19‰$, $n = 23$; IAEA), IAEA-S1 ($\delta^{34}S = -0.43‰ \pm 0.18‰$, $n = 24$), IAEA-S2 ($\delta^{34}S = +21.25‰ \pm 0.28‰$, $n = 24$), IAEA-S3 ($\delta^{34}S = -30.75‰ \pm 5.22‰$, $n = 21$), IAEA-SO-5 ($\delta^{34}S = +0.17‰ \pm 0.24‰$, $n = 24$), Liver 1577b ($\delta^{34}S = +7.62‰ \pm 0.16‰$, $n = 43$; Iso-Analytics), as well as a laboratory protein standard ($\delta^{34}S = +6.25‰ \pm 0.21‰$, $n = 72$; IVA certified).

The extracted bone apatite was measured using a Gasbench II/Delta plus system (Thermo Finnigan) where the samples react with 100% phosphoric acid at 70 °C for 1 h, coupled to the mass spectrometer (Thermo Finnigan Delta Plus; Geo-Center of Northern Bavaria) with a measurement precision of ± 0.1‰ (Doppler et al., 2010). Stable carbon and oxygen isotopic ratios were expressed against the VPDB (Vienna Pee Dee Belemnite; IAEA) standard. The extraction procedure was controlled using laboratory standards, namely NBS 120c ($\delta^{18}O = +21.80‰ \pm 0.56‰$, $n = 14$; National Institute of Standards and Technology NIST, Florida, USA) and NIST 1400 ($\delta^{18}O = +17.01‰ \pm 0.32‰$, $n = 8$; NIST, Florida, USA). The laboratory standard NBS Erl ($\delta^{18}O = +22.55‰ \pm 0.18‰$, $n = 23$; Geo-Center of Northern Bavaria) was used to test for reproducibility and accuracy

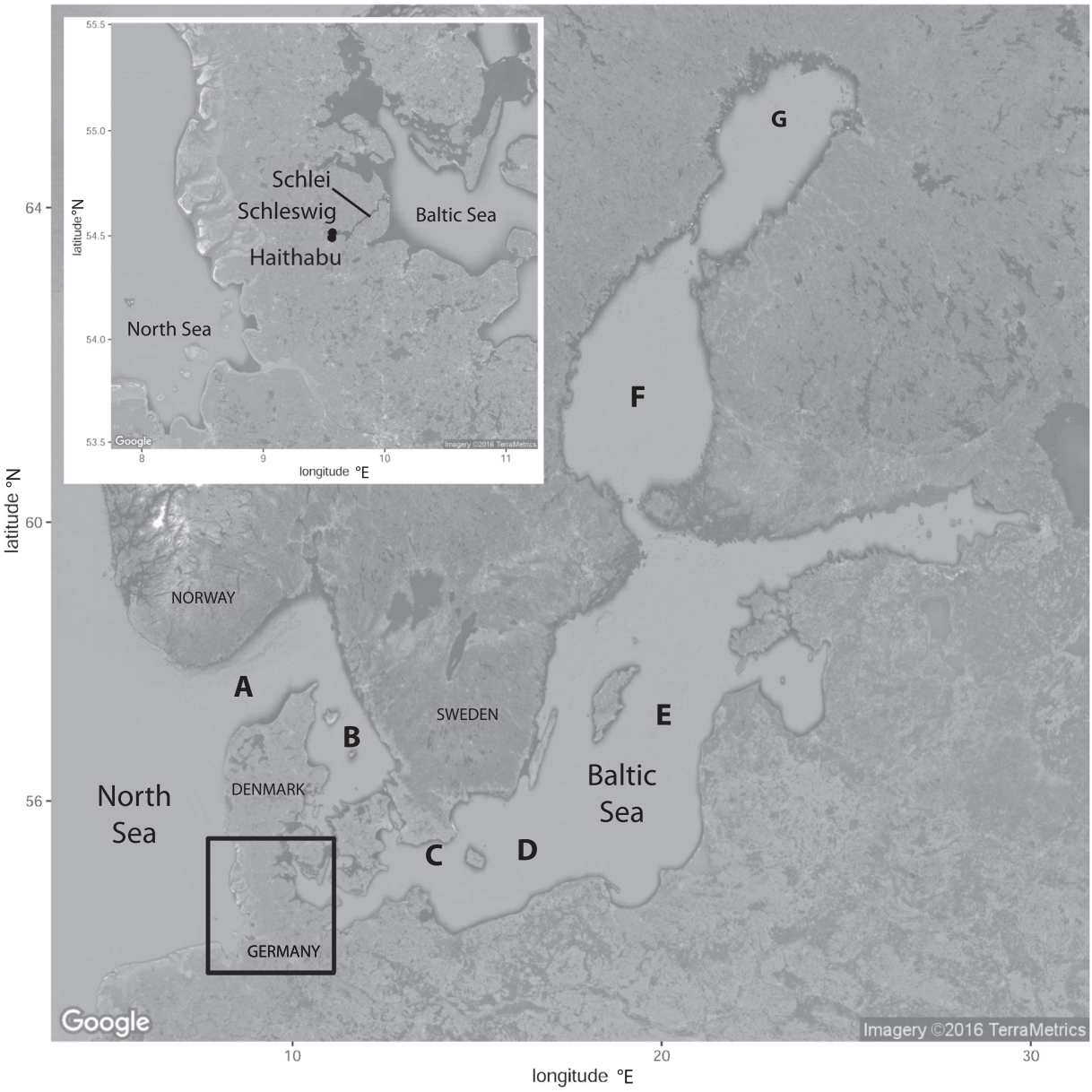


Fig. 1. Map of the area around the Baltic Sea with a more detailed view on the location of the Haithabu and Schleswig sites on the banks of the Schlei inlet. The map was created using the R packages "ggmap" (Kahle and Wickham, 2013) and "ggplot2" (Wickham, 2009); A: Skagerrak, B: Kattegat, C: Arkona Basin, D: Bornholm Basin, E: Gotland Basin, F: Bothnian Sea, G: Bothnian Bay.

of the isotope analyses.

Only samples that met the quality criteria commonly agreed upon were used in our study: A carbon content (%C) around 45%, a nitrogen content (%N) of about 15%, and a sulphur content (%S) in the range between 0.15 and 0.35% are acceptable for mammalian bones (Nehlich and Richards, 2009; von Steinsdorff and Grupe, 2006). Furthermore, the molar C/N ratio should fall within the range between 2.8 and 3.6 (DeNiro, 1985; Harbeck and Grupe, 2009), however it was found that even values up to 4.2 (Harbeck and Grupe, 2009) can be still suitable for stable isotope analysis. Therefore, a C/N ratio of 2.8 to 4.2 was accepted. This led to the inclusion of ten individuals with a ratio of 3.7 and another four with a ratio of 3.8 in the present study. C:S as well as N:S ratios should show values of 600 ± 300 and 200 ± 100, respectively (Nehlich and Richards, 2009). The purity of the bone structural carbonate was verified indirectly according to the relation of $\delta^{13}C_{carb\text{-}coll}$ spacing ($\Delta\delta^{13}C$) and the respective $\delta^{15}N_{coll}$ values (Ambrose et al., 1997; Clementz et al., 2009; Grupe et al., 2009; von Steinsdorff and Grupe, 2006). Comparison of the species-specific trophic levels shown by $\delta^{15}N_{coll}$ and $\Delta\delta^{13}C_{carb\text{-}coll}$ can be used to indirectly check the integrity of bone structural carbonate (von Steinsdorff and Grupe, 2006). Tables S1and S2 include only samples fulfilling all these quality criteria.

For a better resolution of the clustering as well as the sea spray approximation and correction procedure (see below) we round all isotopic values to two decimal places.

2.2. Gaussian Mixture Model clustering

The applicability of Gaussian Mixture Model (GMM) clustering for multi-isotope analyses has already been demonstrated by the example of stable isotope values ($\delta^{13}C_{coll}$, $\delta^{15}N_{coll}$, $\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$) in archaeological fish bones from the Haithabu and Schleswig sites (Göhring et al., 2016). GMM clustering uses all possible combinations of isotopes for the mathematical clustering procedure. Hence, compared to the still common bivariate plotting, this clustering method resulted in a considerable gain of information with regard to palaeobiodiversity studies (Göhring et al., 2016). Therefore, GMM might also be helpful for the

detection of individuals affected by sea spray.GMM was performed with the software "R" (R Core Team, 2016) using the R package "mclust" (Fraley et al., 2012). For more details see Göhring et al. (2016) and references therein.

Due to their different dietary and habitat preferences, terrestrial herbivorous and marine piscivorous mammals should be clearly separated from each other, resulting in two distinct clusters. However, if a sea spray effect was present in at least one of the examined isotopes, one would expect a total number of three resulting clusters, namely a marine cluster, a terrestrial herbivorous cluster, and a mixed cluster falling in between the others. While individuals of the herbivorous cluster would have been affected by sea spray to a smaller degree or even unaffected at all, terrestrial herbivores grouped into the "mixed cluster" should have been markedly affected by sea spray. This "mixed cluster" would include marine mammals exhibiting a higher isotopic freshwater signal, and terrestrial herbivores with a seemingly marine isotopic signal, both a result of the coastal environment. Furthermore, if the sea spray had an influence on a certain isotopic ratio, we would also expect an overlap of terrestrial and marine clusters possibly caused by the sea spray effect. This overlap would then permit an approximation of the sea spray effect (see below).

Discriminant analysis was used for a validation of the resulting GMM models. For this purpose the data were split randomly into two sets, namely a training and a test set with a ratio of 2:1 in order to minimise overfitting (Kohavi, 1995). Semantic labels (e. g. terrestrial, mixed, marine) were not known before clustering and therefore could not be used for discriminant analysis. For this reason, the training set was based on (discrete) labels generated by GMM clustering according to the clustering procedure explained above. This was necessary because of the lack of real labels, which is usually the case in archaeological studies. Especially individuals grouped into the mixed cluster, consequently including terrestrial animals exhibiting isotope values with a "too high" marine signal or marine animals with a "too high" freshwater signal, are difficult to isolate, which was one of the main tasks in our study.

The accuracy of the established model was examined using the training set, while the labelling was validated using the test set. A training error of x indicates an inconsistent labelling of x% of the individuals in the training set clustered using the trained model. Hence, the training error can be understood as the classification error of a model which was trained on the very same data set. On the other hand, a test error of x% describes the proportion of errors in the test set, that has not been clustered with the model before, while fitting it into the trained model. Consequently, x% of the data, which was not used to train the model, was incorrectly classified. This procedure was repeated 100 times to calculate a mean training and test error, respectively, as well as the corresponding standard deviation.

2.3. Approximation of the sea spray effect

The approximation procedure was largely based on the simplifying assumption that terrestrial herbivorous mammals should exhibit more negative isotopic ratios in all isotopic dimensions compared to marine piscivorous mammals leading to two distinct groups. This assumption is based on the fact that, in case of coastal environments, marine systems (including all organic and inorganic components) are usually enriched in ^{13}C, ^{15}N, ^{18}O, and ^{34}S compared to the corresponding terrestrial C_3 environment (see e. g. Clementz et al., 2006; Clementz and Koch, 2001; Fry and Sherr, 1984; Hobson, 1999; Koch, 2007; Miyake and Wada, 1967; Mook, 2000; Mook, 1971; Nehlich, 2015; Richards et al., 2006; Richards and Hedges, 1999; Schaffner and Swart, 1991; Schoeninger and DeNiro, 1984). This can be explained as follows: Marine organisms are enriched in both collagen ^{13}C and ^{15}N due to longer food chains in the marine environment and thus a higher level of trophic enrichment (Miyake and Wada, 1967). Carbon isotope values differ between terrestrial and marine environments mainly due to the different isotopic signature of plants and plant suspended decomposition products, i.e. particulate organic carbon (POC) (Schaffner and Swart, 1991). Evaporation of ocean water leads to a fractionation of oxygen isotopes due to the preferential evaporation of the lighter ^{16}O compared to ^{18}O (Craig and Gordon, 1965). Hence, ocean water becomes more enriched in ^{18}O. Differences in e. g. drinking water, food, and ambient temperature will cause a difference between marine and terrestrial animals. Sulphur input to terrestrial systems is primarily caused by precipitation sulphate with a $\delta^{34}S$ value of 2 to 8‰. Marine systems, on the other hand, contain marine sulphate with a relatively constant sulphur isotope value of about 21‰ (Peterson and Howarth, 1987; Rees et al., 1978).

The approximation method is explained stepwise in the results' section. It should be emphasised that all calculations and thresholds used in this study are site-specific. They can probably not be directly applied to data sets from other sites since the sea spray effect differs along the Baltic coast because of varying salinity levels (see above). Furthermore, our data set probably also included some imported animals. As these animals were however not clearly distinguishable from local ones according to their isotopic signature, we cannot exclude some inaccuracy in the approximation procedure. Nevertheless, we suggest a general approximation method that can be adapted to other data sets.

3. Results

3.1. Gaussian Mixture Model clustering with uncorrected isotope values

According to our expectations of three clusters (terrestrial, mixed, marine; see above) the number of clusters was a priori chosen. Analysing the four-dimensional data set, the model mathematically preferred by "mclust" was "VEI", which allows for diagonal clusters of equal shape (Fraley et al., 2012). This model also showed the most distinct results. In the case of three clusters, 38 individual animals were grouped into cluster 1, and 25 into both the second and third cluster (Table 1). Mean training and test error were 0.98% ± 1.13% and 5.80% ± 4.51%, respectively.

Comparing clusters 1 and 2 with each other, a small difference was detectable in all isotopic ratios with the exception of $\delta^{18}O_{carb}$ (Fig. 2, Table 2). Animals grouped into the first cluster exhibited slightly more positive isotopic signatures compared to animals of the second cluster. This indicates a higher marine influence on the animals of cluster 1 compared to individuals of the second cluster. With the exception of two individuals (common seal Hb23 and sperm whale Hb50), only terrestrial herbivores were grouped into the first cluster. The second cluster contained terrestrial herbivores only, which according to their isotopic values were probably not or only slightly affected by sea spray (Table S2).

Cluster 3 was markedly different from the other ones with regard to $\delta^{13}C_{coll}$ and $\delta^{15}N_{coll}$ values. Mammals of cluster 3 were enriched with both ^{13}C and ^{15}N in the collagen fraction indicative of a marine signal in the dietary protein (Fig. 2, Table 2). Also the mean $\delta^{13}C_{carb}$ value of − 9.88‰ (± 1.75‰) was more positive than the respective means of clusters 1 and 2, respectively (Table 2). However, measured $\delta^{18}O_{carb}$ values did not exhibit such a difference (Fig. 2, Table 2). These results were in agreement with the fact that only marine mammals were grouped into the third cluster. An overlap between all clusters according to their mean stable oxygen isotope value was clearly evident (Table 2) contradicting the fact that marine individuals should usually be enriched with ^{18}O compared to terrestrial ones (Hobson, 1999).

Performing GMM on the five-dimensional data set including $\delta^{34}S_{coll}$ values (n = 22) produced quite similar results. For the a priori chosen number of three clusters, GMM again obtained one cluster including herbivores only (cluster 2, n = 7), one cluster with only marine mammals (cluster 3, n = 7), and a "mixed" cluster (cluster 1, n = 8) including herbivores with an intermediate isotopic signature and thus a

Table 1
GMM clustering results performed with three clusters and the "VEI" model according to the uncorrected four-dimensional data set.

Cluster 1	Cluster 2	Cluster 3
Aurochs Hb47	Hare Hb5	Common seal Hb24
Aurochs Hb48	Hare S25	Common seal Hb25
Cattle Hb53	Hare S26	Common seal Hb26
Cattle Hb54	Hare S27	Common seal Hb27
Cattle Hb55	Hare S28	Common seal KHb30
Cattle Hb56	Horse Hb88	Common seal KHb31
Cattle Hb57	Horse Hb89	Common seal 27SH1Ph
Cattle Hb58	Horse Hb90	Common seal 28SH2R
Cattle Hb59	Horse Hb92	Common seal 29SH3V
Cattle Hb60	Red deer Hb36	Grey seal S17
Cattle Hb61	Red deer Hb38	Grey seal 20KR1Ph
Cattle Hb62	Red deer S1	Grey seal 21KR2T
Common seal Hb23	Red deer S2	Grey seal 22KR3S
Fallow deer S6	Red deer S3	Grey seal 23KR4V
Hare Hb1	Red deer S4	Grey seal 24KR5Mt
Hare Hb2	Red deer S5	Grey seal 71KR1BuTy
Hare Hb3	Roe deer Hb40	Grey seal 72KR2Sc
Hare S24	Roe deer Hb41	Grey seal 73KR3Mt
Horse Hb83	Roe deer Hb42	Grey seal 74KR4Fi
Horse Hb84	Roe deer Hb43	Grey seal 75KR5Mt
Horse Hb85	Roe deer S8	Ringed seal Hb28
Horse Hb86	Roe deer S9	Saddleback seal Hb30
Horse Hb87	Roe deer S10	Saddleback seal KHb25
Horse Hb91	Roe deer S12	Saddleback seal 26SR2Fi
Moose Hb46	Squirrel Hb7	Walrus KS76
Rabbit S29		
Rabbit S30		
Red deer Hb35		
Red deer Hb39		
Roe deer S11		
Sheep Hb73		
Sheep Hb74		
Sheep Hb75		
Sheep Hb76		
Sheep Hb77		
Sheep Hb78		
Sheep Hb79		
Sperm whale Hb50		

marine-like signal (Fig. 3, Table 3). Discriminant analysis showed training and test errors of 0.0% ± 0.0% and 21.4% ± 16.2%, respectively.

The calculated mean isotopic ratios of the respective clusters were very similar to the means in the four-dimensional data set (Table 4). Mean $\delta^{18}O_{carb}$ values of cluster 1 and 3 showed an overlap, again contradicting the expectations.

3.2. Approximation of the minimum sea spray effect

The effect leading to an overlap of the different groups (Figs. 2 and 3), which could not be explained by the actual diet and habitat of the different animals, was approximated. According to our basic assumption (see above) marine mammals should be clearly separated from terrestrial herbivorous mammals in all isotopic dimensions. The first task was the choice of appropriate isotopic measurements for the definition of marine and terrestrial isotopic boundaries.

Considering $\delta^{13}C_{coll}$ or $\delta^{15}N_{coll}$ values only, common seal Hb23 and sperm whale Hb50 were more depleted in ^{13}C and ^{15}N than several terrestrial herbivores under study (Tables S3 and S4). These individuals were grouped into the mixed cluster 2 (Table 1) accordingly, indicating a relatively high freshwater input (Mook, 1971) or origin from northern (colder) latitudes (Marshall et al., 2007) in the case of the sperm whale. Therefore, these specimens could not be chosen as exclusively marine individuals necessary for our approximations. Both species should not be chosen for the definition of an isotopic marine limit for another reason. Common seals often swim up the rivers (Reichstein, 1991) which will introduce a freshwater isotopic signal into their tissues as especially detected for seal Hb23. Sperm whales were not local to the Baltic sea but rather originated from much further up north (Heinrich, 1991). Accordingly, as illustrated by Figs. 2 and 3, no distinct overlap of the terrestrial and marine cluster was detectable with regard to the carbon and nitrogen stable isotope values of the collagen fraction. Consequently, although we could not exclude a sea spray effect present biasing these isotopic signatures, we were not able to approximate the effect in neither $\delta^{13}C_{coll}$ nor $\delta^{15}N_{coll}$.

However, there was a clear overlap detectable between terrestrial and marine clusters in the carbonate fraction ($\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$) (Figs. 2 and 3, Tables S5 and S6). Herbivores with a marine-like isotopic signature even exceeding those of some marine mammals, could be used for an approximation of the sea spray effect (Göhring et al., 2015).

The approximation of the sea spray effect was performed on the four-dimensional data set (n = 138) in a similar way as in the case of $\delta^{34}S_{coll}$ (Göhring et al., 2015) with one exception, namely the additional calculation of the marine impact (see Section 3.3):

Following our underlying assumption (see above), we had to define a lower marine limit equal to the marine piscivorous mammal with the lowest respective isotopic value. The lower marine limit was set to − 12.96‰ ($\delta^{13}C_{carb}$; grey seal 23KR4V) and − 10.21‰ ($\delta^{18}O_{carb}$; grey seal 72KR2Sc), respectively. For $\delta^{34}S_{coll}$, grey seal S17 with a sulphur isotopic ratio of + 12.56‰ was chosen as lower marine limit (Göhring et al., 2015). Grey seals once were probably local at the Baltic coast (Reichstein, 1991). Common seal KHb31, saddleback seal 26SR2Fi, as well as grey seals 73KR3Mt and 71KR1BuTy were consciously not chosen as marine limit (although they exhibited even more negative isotopic values) since the density curve of the $\delta^{18}O_{carb}$ values of the examined marine mammals exhibited a smaller second peak (Fig. S1), probably indicating non-local origin or a different habitat and diet choice, respectively. For this reason, grey seal 73KR3Mt was not chosen as lower marine limit for $\delta^{13}C_{carb}$, too, although its isotopic ratio (− 13.59‰) was within a normal range (Fig. S1). Sperm whale Hb50 ($\delta^{13}C_{carb}$ = − 14.48‰) was not chosen as limit as this individual was also not local (see above).

The difference (Δ) between the chosen marine limit and those herbivores with an isotope value higher than the lower marine limit were calculated and converted into percentage values (Δ [%]) according to the following equations (Göhring et al., 2015):

$$\Delta = \text{isotope value} - \text{lower marine limit} \quad (1)$$

$$\Delta\,[\%] = \frac{\Delta * 100}{\text{isotope value}} \quad (2)$$

For the approximation of the minimum sea spray effect the percentage difference between the terrestrial herbivore exhibiting the most positive and thus most marine isotope value and the lower marine limit was calculated. As one would expect a decrease of the sea spray effect with increasing distance from the coast as found for sulphur stable isotopes (Mizota and Sasaki, 1996; Wakshal and Nielsen, 1982; Zazzo et al., 2011), as well as a probably different influence of the effect on the particular individuals examined in our study, e. g. depending on the amount of affected food in the diet, we preferred to use the percentage difference. Due to the correction according to the respective percentage difference individuals not or only slightly affected by sea spray, and thus exhibiting rather low isotope values, would be corrected less, while individuals influenced by sea spray and exhibiting rather high isotope values would be corrected more (Göhring et al., 2015).

Accordingly, the largest percentage difference was calculated as 32.8% for carbonate carbon with an isotopic value of − 9.76‰ evident in a hare (Hb3) and 62.8% for carbonate oxygen (hare S28, $\delta^{18}O_{carb}$ = − 6.27‰), respectively. Interestingly, hare Hb3 also showed the highest sulphur isotopic ratio (+ 15.05‰) leading to a percentage difference of 16.5% in sulphur isotopes (Göhring et al., 2015).

These values (32.8%, 62.8%, as well as 16.5%) can be understood as

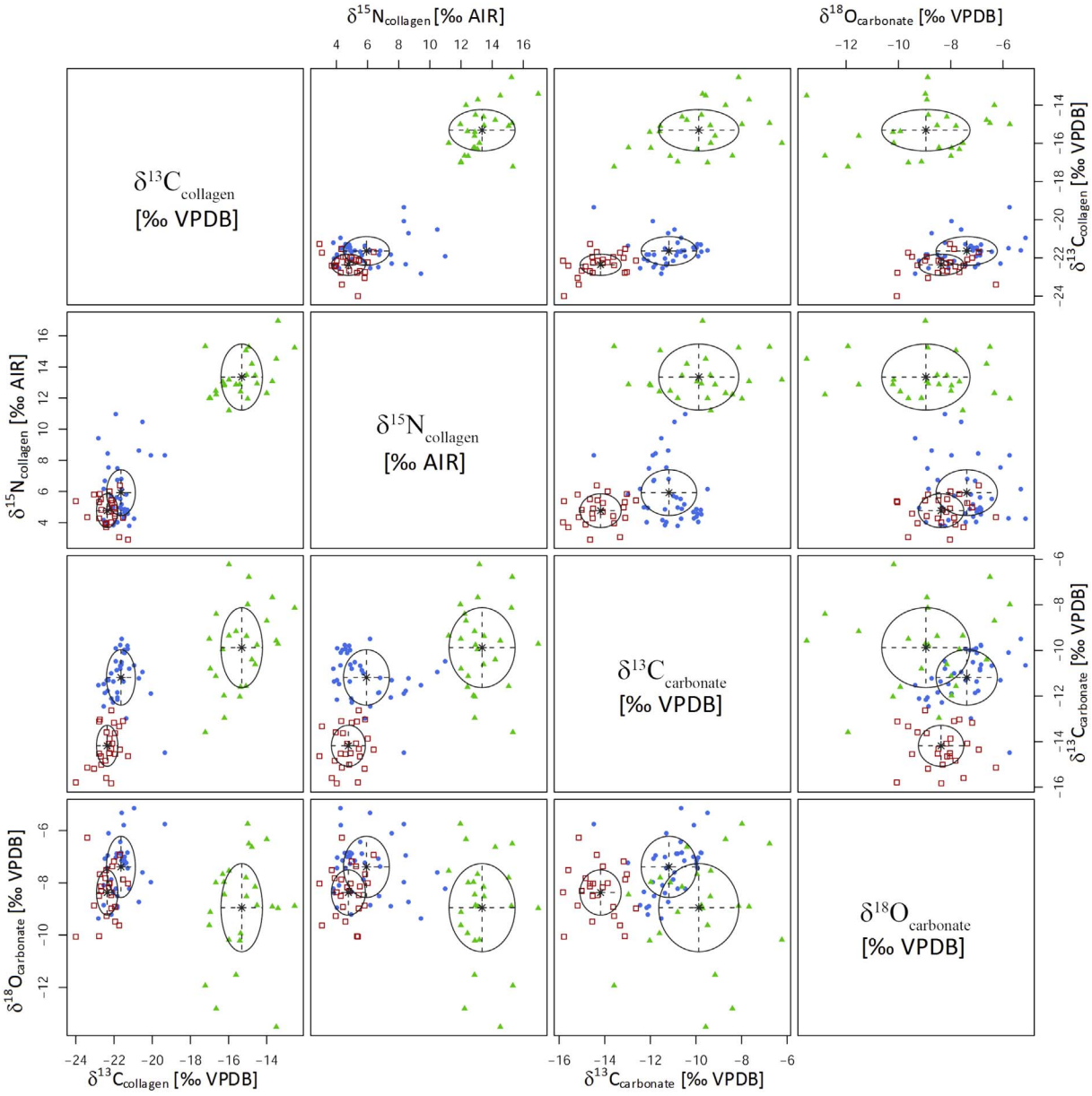


Fig. 2. Pairwise bivariate plot of the resulting three clusters performed with the "VEI" model (Fraley et al., 2012) on the uncorrected four-dimensional ($\delta^{13}C_{coll}$, $\delta^{15}N_{coll}$, $\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$) data set; cluster 1: dots, cluster 2: squares, cluster 3: triangles.

Table 2
Mean and standard deviation (sd) of the uncorrected four-dimensional data set according to GMM clustering performed with three clusters.

	Cluster 1	Cluster 2	Cluster 3
Mean			
$\delta^{13}C_{coll}$ [‰ VPDB]	− 21.64	− 22.36	− 15.31
$\delta^{15}N_{coll}$ [‰ AIR]	+ 5.93	+ 4.77	+ 13.35
$\delta^{13}C_{carb}$ [‰ VPDB]	− 11.18	− 14.18	− 9.88
$\delta^{18}O_{carb}$ [‰ VPDB]	− 7.39	− 8.37	− 8.95
sd			
$\delta^{13}C_{coll}$ [‰ VPDB]	0.75	0.56	1.09
$\delta^{15}N_{coll}$ [‰ AIR]	1.47	1.09	2.12
$\delta^{13}C_{carb}$ [‰ VPDB]	1.22	0.90	1.75
$\delta^{18}O_{carb}$ [‰ VPDB]	1.17	0.87	1.69

the minimum sea spray effect on the isotopic ratios of these hares. All other terrestrial animals, including herbivores, carnivores, omnivores, as well as freshwater animals, had at least to be corrected for this minimum effect under the assumption that there should be no overlap between terrestrial herbivorous and marine piscivorous mammals (see above).

3.3. Correction for the minimum sea spray effect

The correction of measured isotopic signatures for the minimum sea spray effect can be conducted similar to the procedure performed previously with stable sulphur isotope values (Göhring et al., 2015).

However, due to the presence of negative isotopic ratios in our data set, unlike positive $\delta^{34}S_{coll}$ values in the previous study, we had to introduce an additional parameter, called "marine impact". This parameter is necessary because without considering it, we would correct more negative values to a higher extent than less negative ones due to their higher modulus, whereas more negative isotopic values should be

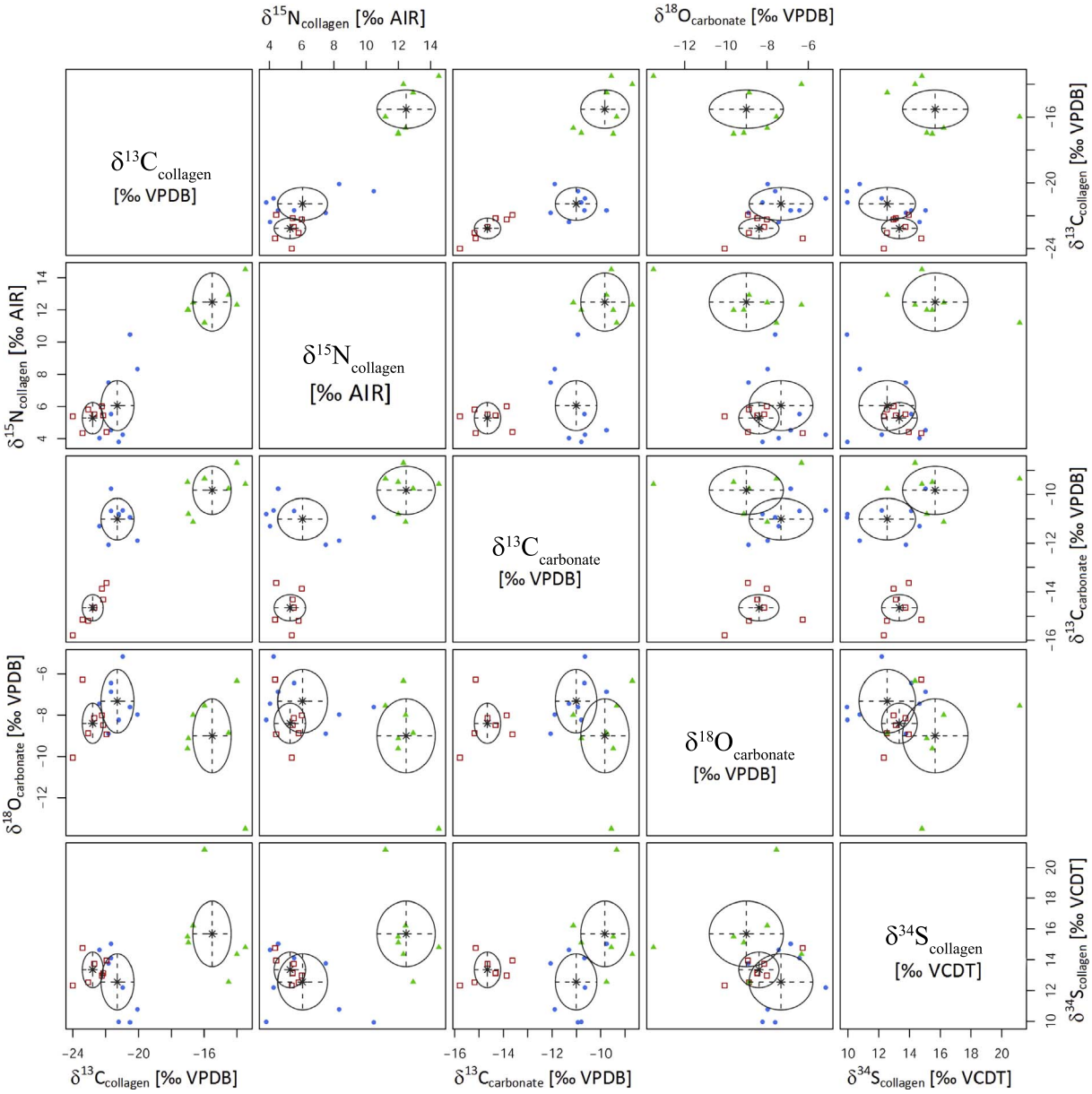


Fig. 3. Pairwise bivariate plot of the resulting three clusters performed with the "VEI" model (Fraley et al., 2012) on the uncorrected five-dimensional ($\delta^{13}C_{coll}$, $\delta^{15}N_{coll}$, $\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$, $\delta^{34}S_{coll}$) data set; cluster 1: dots, cluster 2: squares, cluster 3: triangles.

Table 3
GMM clustering results performed with three clusters and the "VEI" model according to the uncorrected five-dimensional data set.

Cluster 1	Cluster 2	Cluster 3
Aurochs Hb47	Hare S28	Common seal Hb25
Cattle Hb57	Horse Hb92	Common seal KHb30
Hare Hb2	Red deer S3	Common seal KHb31
Hare Hb3	Red deer S4	Grey seal S17
Hare S24	Red deer S5	Saddleback seal Hb30
Roe deer S11	Roe deer Hb40	Saddleback seal KHb25
Sheep Hb74	Roe deer Hb41	Walrus KS76
Sheep Hb79		

Table 4
Mean and standard deviation (sd) of the uncorrected five-dimensional data set according to GMM clustering performed with three clusters.

	Cluster 1	Cluster 2	Cluster 3
Mean			
$\delta^{13}C_{coll}$ [‰ VPDB]	− 21.28	− 22.78	− 15.52
$\delta^{15}N_{coll}$ [‰ AIR]	+ 6.06	+ 5.27	+ 12.49
$\delta^{13}C_{carb}$ [‰ VPDB]	− 11.01	− 14.66	− 9.83
$\delta^{18}O_{carb}$ [‰ VPDB]	− 7.32	− 8.39	− 9.00
$\delta^{34}S_{coll}$ [‰ VCDT]	+ 12.56	+ 13.34	+ 15.67
sd			
$\delta^{13}C_{coll}$ [‰ VPDB]	1.00	0.63	1.17
$\delta^{15}N_{coll}$ [‰ AIR]	1.55	0.98	1.81
$\delta^{13}C_{carb}$ [‰ VPDB]	0.85	0.54	0.99
$\delta^{18}O_{carb}$ [‰ VPDB]	1.54	0.97	1.80
$\delta^{34}S_{coll}$ [‰ VCDT]	1.83	1.16	2.14

corrected to a lesser extent as they are possibly less or not affected by sea spray at all. Consequently, we had to rearrange the original correction equation

$$\delta^{Y}X_{corrected} = \delta^{Y}X - |\Delta\,[\%] * \delta^{Y}X|$$

(after Göhring et al., 2015) (3)

into the modified equation

$$\delta^{Y}X_{corrected} = \delta^{Y}X - |m * \Delta\,[\%] * \delta^{Y}X| \quad (4)$$

with stable isotopic ratio $\delta^{Y}X$, approximated sea spray effect Δ (in percent), and marine impact m (see below).

The marine impact (m) of each individual animal could be calculated using the two most extreme isotopic values exhibited by terrestrial herbivorous mammals (see Eqs. 5–7). It is important to emphasise that this is not completely identical with the calculation of the terrestrial and marine input in the mixed diet as calculated for $\delta^{34}S_{coll}$ values by Göhring et al. (2015) as a demonstration of the influence of the sea spray effect calculated with the isotope values of the most obviously terrestrial herbivorous mammal and the most obviously marine and piscivorous mammal, respectively. In contrast to the terrestrial and marine input, respectively, we wanted to define the relative position of each individual animal within the range of the two most extreme herbivores with a 100% marine and a 100% terrestrial signal, identical to the individuals required to be corrected most and least, respectively.

$\delta^{13}C_{carb}$ values measured in terrestrial herbivores from Haithabu and Schleswig ranged from − 15.83‰ (hare Hb5) to − 9.76‰ (hare Hb3). Therefore, the marine impact (m), as well as the corresponding terrestrial impact (t), of the different animals could be calculated by a linear mixing model as follows:

$$\delta^{13}C_{carb} = -15.83 * t - 9.76 * m \text{ with } t + m = 1. \quad (5)$$

Due to the choice of these two most extreme herbivorous values several other, however not herbivorous, terrestrial animals exhibited $\delta^{13}C_{carb}$ values higher or lower than these most extreme values (see Table S5). Logically, calculation of the marine impact in these cases led to results lower than 0 and higher than 1, respectively. The possible influence and the *caveat* of those values is discussed later.

The isotopic range found in $\delta^{18}O_{carb}$ values of terrestrial herbivorous mammals was between − 10.06‰ (roe deer Hb40) and − 6.27‰ (hare S28). Similar to the calculation for $\delta^{13}C_{carb}$ we could again use a linear mixing model:

$$\delta^{18}O_{carb} = -10.06 * t - 6.27 * m \text{ with } t + m = 1. \quad (6)$$

Again, several terrestrial mammals showed a marine impact higher or lower than the two most extreme herbivores (Table S6; see below).

In the case of $\delta^{34}S_{coll}$ the two most extreme herbivores, aurochs Hb47 (+ 9.94‰) and hare Hb3 (+ 15.05‰), were used for calculation of the marine impact according to the equation

$$\delta^{34}S_{coll} = 9.94 * t + 15.05 * m \text{ with } t + m = 1. \quad (7)$$

A single specimen, pig Hb65 ($\delta^{34}S_{coll}$ = + 16.23‰), showed a marine impact higher than 1 (Table S6; see below).

After correction for the minimum sea spray effect according to Eq. (4) in $\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$, and $\delta^{34}S_{coll}$, respectively, only few animals (mathematically) remained above the lower marine limit (Fig. S2, Tables S5–S7):

Three mammals (red fox S18, sheep Hb77, common otter 32FO1Hepi) exhibited $\delta^{13}C_{carb}$ ratios above − 12.96‰, the lower marine limit (grey seal 23KR4V), with corrected isotopic ratios of − 12.81‰, − 12.74‰, and − 10.06‰, respectively (Fig. S2, Table S5).

In the case of $\delta^{18}O_{carb}$, after correction for the minimum sea spray effect seven individuals exhibited still "too marine" isotopic values (> − 10.21‰, grey seal 72KR2Sc): one hare (S28), two red foxes (S18, S21), and four sheep (Hb76, Hb77, Hb78, Hb79). Their corrected isotopic values were in the range between − 10.21‰ and − 8.98‰ (Fig. S2, Table S6).

Only pig Hb65 with a corrected $\delta^{34}S_{coll}$ ratio of + 12.93‰ remained above the chosen lower marine limit of + 12.56‰ (grey seal S17), what was also the case with the previous correction equation of Göhring et al. (2015) resulting in a corrected sulphur isotope value of + 13.55‰ for the respective pig (Fig. S2, Table S7).

As mentioned before, some individual isotopic signatures were obviously "overcorrected". This was caused by a marine impact smaller than 0 or higher than 1, respectively. A negative marine impact was present in five animals, namely common otter Hb20 ($\delta^{13}C_{carb}$ = − 16.26‰), beaver Hb9 ($\delta^{13}C_{carb}$ = − 16.06‰), cat Hb104 ($\delta^{18}O_{carb}$ = − 10.64‰), pig Hb70 ($\delta^{18}O_{carb}$ = − 10.18‰), as well as red fox S22 ($\delta^{34}S_{coll}$ = + 8.15‰). Marine signals higher than 1 were calculated for the eight individuals red fox S18 ($\delta^{13}C_{carb}$ = − 9.75‰, $\delta^{18}O_{carb}$ = − 6.20‰), sheep Hb77 ($\delta^{13}C_{carb}$ = − 9.49‰, $\delta^{18}O_{carb}$ = − 5.52‰), common otter 32FO1Hepi ($\delta^{13}C_{carb}$ = − 6.75‰), sheep Hb78 ($\delta^{18}O_{carb}$ = − 6.10‰), sheep Hb76 ($\delta^{18}O_{carb}$ = − 5.79‰), sheep Hb79 ($\delta^{18}O_{carb}$ = − 5.14‰), red fox S21 ($\delta^{18}O_{carb}$ = − 4.80‰), as well as pig Hb65 ($\delta^{34}S_{coll}$ = + 16.23‰) (Fig. S2, Tables S5–S7). The aberration in the correction process was particularly noticeable for animals with a negative marine signal, indicating an isotopic signature more negative than the respective herbivore with the most negative isotope value. Although according to our assumption animals exhibiting more negative isotope values should be corrected less, these certain individuals were corrected more than expected (and desired) due to the modulus in Eq. (4), clearly evident in Fig. S2 in the form of a larger distance between the respective uncorrected and corrected isotopic values.

3.4. Gaussian Mixture Model clustering with corrected carbonate and sulphur isotope values

Performing GMM on the four-dimensional data set with carbonate isotopic values corrected for sea spray (see above) led to a better separation of all three clusters in the carbonate fraction, while the separation with regard to the collagen isotopes ($\delta^{13}C_{coll}$, $\delta^{15}N_{coll}$) remained unmodified as both dimensions were not corrected at all as explained above (Fig. 4). Due to the correction for the sea spray effect terrestrial mammals of both cluster 1 and 2 were shifted towards more negative values in the carbonate fraction. Training and test error for this model were 0.48% ± 0.92% and 7.90% ± 5.25%, respectively.

After correction for the sea spray effect, the nine mammals that remained in the first cluster (Table 5) at the same time exhibited the most extreme marine values within the examined terrestrial mammals (Table 6). Common seal Hb23 was still caught by the first cluster. The isotopic values of this animal, as well as all other marine mammals, remained unchanged. All other herbivorous mammals previously grouped into cluster 1 (Table 1) were now grouped into the second cluster (Table 5). After correction for the sea spray effect 53 instead of previously 25 mammals were grouped into cluster 2. Cluster 3 that contained all marine mammals was changed only slightly, and now also included sperm whale Hb50 (Table 5), that had previously been grouped into the first cluster (see above).

After correction for the minimum sea spray effect in $\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$, and $\delta^{34}S_{coll}$ simultaneously (training error: 0.00% ± 0.00%, test error: 17.1% ± 13.2%), only two mammals remained in the first cluster, namely aurochs Hb47 and roe deer S11. All other herbivores previously grouped into cluster 1 (Table 3) were now found in the second cluster. The marine cluster (cluster 3) was not changed at all (Fig. 5, Tables 7 and 8).

3.5. Gaussian Mixture Model clustering with corrected carbonate and nitrogen isotope values

Even after correction for the minimum sea spray effect individuals

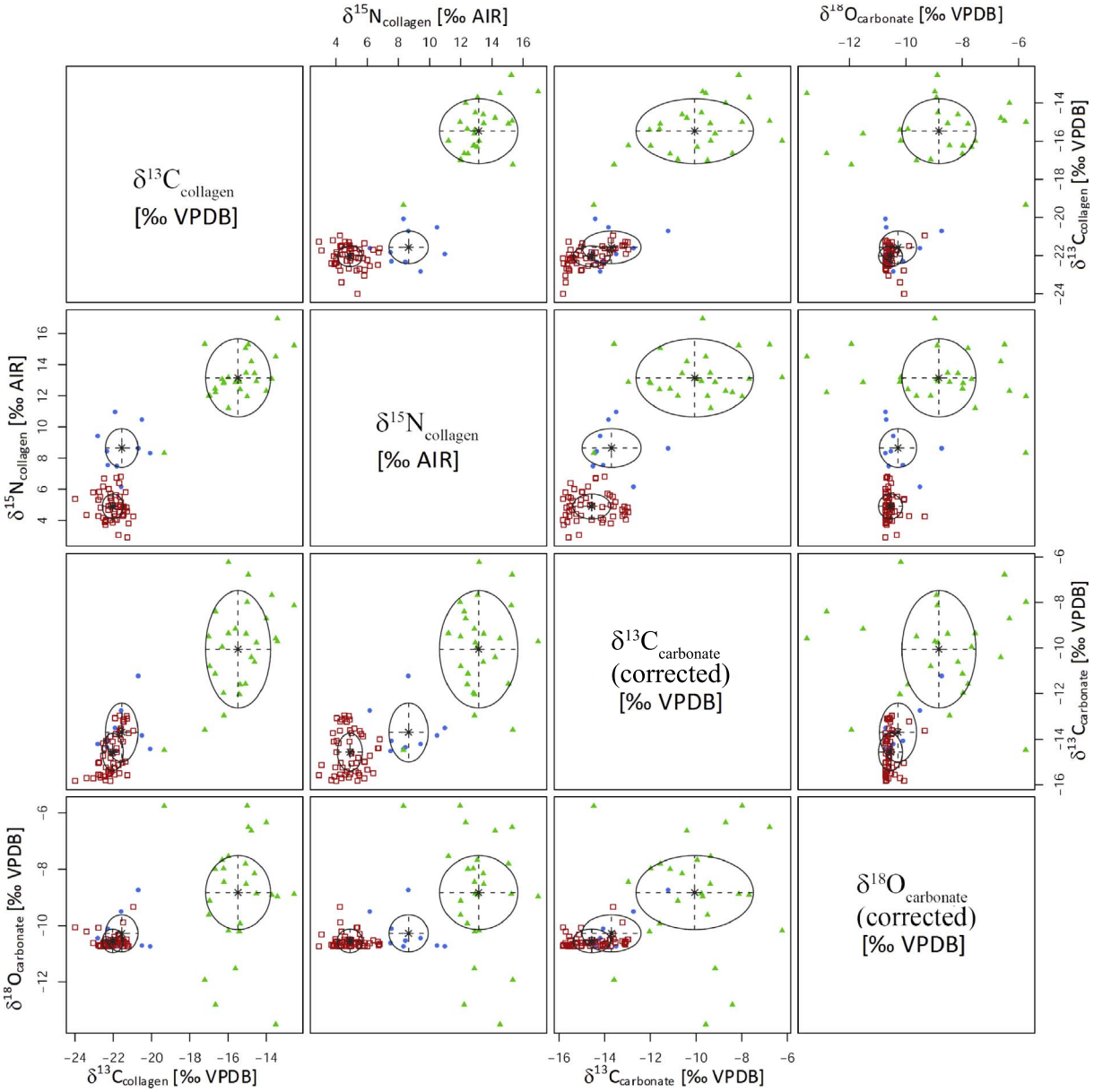


Fig. 4. Pairwise bivariate plot of the resulting three clusters performed with the "VEI" model (Fraley et al., 2012) on the four-dimensional data set (terrestrial mammals corrected for sea spray in $\delta^{13}C_{carb}$ (32.8%) and $\delta^{18}O_{carb}$ (62.8%)); cluster 1: dots, cluster 2: squares, cluster 3: triangles.

grouped into the mixed cluster still showed unusually positive $\delta^{15}N_{coll}$ values (Fig. 4). As shown by Göhring et al. (2015), a limnic influence seems to have a distinct effect on nitrogen stable isotopes leading to an enrichment in ^{15}N. This effect is probably triggered by aquatic insects as insects serve as both direct food source and plant fertiliser after their death (Gratton et al., 2008). This results in a shift of nitrogen stable isotope values of terrestrial vertebrates towards higher values. Consequently, correction for both the minimum sea spray effect in the carbonate isotopes and the limnic influence on the nitrogen isotopic values (20.8%; Göhring et al., 2015) should mathematically eliminate the mixed cluster.

If we forced the data only corrected for the minimum sea spray effect into two clusters, GMM produced one cluster including herbivores only and a second cluster including all marine mammals as well as aurochs Hb47 and rabbit S30, the two herbivores of our study exhibiting the most enriched nitrogen stable isotope values of + 10.48‰ and + 10.98‰, respectively (Fig. S3, Tables S4, S8 and S9). Discriminant analysis showed mean training and test errors of 0.52% ± 0.90% and 2.57% ± 2.84%, respectively. However, additional correction for the approximated limnic influence of 20.8% (Göhring et al., 2015) led to a pure terrestrial as well as a pure marine cluster (Fig. S4, Tables S10 and S11) with a mean training error of 0.02% (± 0.17%) as well as a mean test error of 0.80% (± 2.07%).

4. Discussion

GMM is able to detect terrestrial herbivorous mammals possibly affected by sea spray during their lifetime. Due to the enrichment of their isotopic signature these animals are grouped into a separated cluster located between terrestrial herbivores with more negative isotopic signatures (not or only slightly affected by the sea spray), and marine mammals which exhibit more positive isotopic values due to their marine habitat and piscivorous diet. As terrestrial herbivores should not exhibit isotopic values higher than local marine mammals from the same site (see above), shifts due to sea spray can be detected as an overlap of terrestrial herbivorous and marine mammals. However, as explained before, if no overlap is present in particular isotopic systems, such as $\delta^{13}C_{coll}$ and $\delta^{15}N_{coll}$ in the present study, this does not

Table 5
GMM clustering results performed with three clusters and the "VEI" model according to the four-dimensional data set corrected for the sea spray effect.

Cluster 1	Cluster 2		Cluster 3
Aurochs Hb47	Aurochs Hb48	Moose Hb46	Common seal Hb24
Common seal Hb23	Cattle Hb53	Red deer Hb35	Common seal Hb25
Fallow deer S6	Cattle Hb54	Red deer Hb36	Common seal Hb26
Rabbit S29	Cattle Hb55	Red deer Hb38	Common seal Hb27
Rabbit S30	Cattle Hb56	Red deer Hb39	Common seal KHb30
Roe deer S11	Cattle Hb57	Red deer S1	Common seal KHb31
Sheep Hb74	Cattle Hb58	Red deer S2	Common seal 27SH1Ph
Sheep Hb77	Cattle Hb59	Red deer S3	Common seal 28SH2R
Sheep Hb78	Cattle Hb60	Red deer S4	Common seal 29SH3V
	Cattle Hb61	Red deer S5	Grey seal S17
	Cattle Hb62	Roe deer Hb40	Grey seal 20KR1Ph
	Hare Hb1	Roe deer Hb41	Grey seal 21KR2T
	Hare Hb2	Roe deer Hb42	Grey seal 22KR3S
	Hare Hb3	Roe deer Hb43	Grey seal 23KR4V
	Hare Hb5	Roe deer S8	Grey seal 24KR5Mt
	Hare S24	Roe deer S9	Grey seal 71KR1BuTy
	Hare S25	Roe deer S10	Grey seal 72KR2Sc
	Hare S26	Roe deer S12	Grey seal 73KR3Mt
	Hare S27	Sheep Hb73	Grey seal 74KR4Fi
	Hare S28	Sheep Hb75	Grey seal 75KR5Mt
	Horse Hb83	Sheep Hb76	Ringed seal Hb28
	Horse Hb84	Sheep Hb79	Saddleback seal Hb30
	Horse Hb85	Squirrel Hb7	Saddleback seal KHb25
	Horse Hb86		Saddleback seal 26SR2Fi
	Horse Hb87		Sperm whale Hb50
	Horse Hb88		Walrus KS76
	Horse Hb89		
	Horse Hb90		
	Horse Hb91		
	Horse Hb92		

Terrestrial mammals were corrected for sea spray in $\delta^{13}C_{carb}$ (32.8%) and $\delta^{18}O_{carb}$ (62.8%).

Table 6
Mean and standard deviation (sd) of the corrected four-dimensional data set according to GMM clustering performed with three clusters.

	Cluster 1	Cluster 2	Cluster 3
Mean			
$\delta^{13}C_{coll}$ [‰ VPDB]	− 21.56	− 22.03	− 15.48
$\delta^{15}N_{coll}$ [‰ AIR]	+ 8.65	+ 4.90	+ 13.15
$\delta^{13}C_{carb}$ [‰ VPDB]	− 13.70	− 14.57	− 10.06
$\delta^{18}O_{carb}$ [‰ VPDB]	− 10.27	− 10.54	− 8.83
sd			
$\delta^{13}C_{coll}$ [‰ VPDB]	0.85	0.55	1.70
$\delta^{15}N_{coll}$ [‰ AIR]	1.26	0.81	2.51
$\delta^{13}C_{carb}$ [‰ VPDB]	1.29	0.83	2.58
$\delta^{18}O_{carb}$ [‰ VPDB]	0.66	1.09	1.32

Terrestrial mammals were corrected for sea spray in $\delta^{13}C_{carb}$ (32.8%) and $\delta^{18}O_{carb}$ (62.8%).

necessarily imply absence of a sea spray effect in these isotopes. The absence of a quantifiable sea spray effect in collagen nitrogen stable isotopes contrasts the results of Richards et al. (2006) who assumed a sea spray effect in nitrogen stable isotopes of sheep/goats, illustrating how site-specific the sea spray can be. However, even in this case, there was no overlap detectable between sheep/goats and marine mammals (seals) with a mean stable nitrogen isotope value of + 7.0‰ ± 0.9‰ and + 17.2‰ ± 1.5‰ (Richards et al., 2006), respectively. Furthermore, compared to the $\delta^{15}N_{coll}$ values of e. g. fallow deer S6 (+ 9.43‰), aurochs Hb47 (+ 10.48‰), and rabbit S30 (+ 10.98‰) in our study, a mean nitrogen isotope value of about + 7‰ seems to be common. Nevertheless, we cannot exclude that these values were affected by sea spray effect. On the other hand, also a limnic influence might have caused the enrichment of $\delta^{15}N_{coll}$ of the sheep or goats in the study by Richards et al. (2006), what is probably the case for our data. Moreover, as both sea spray effect and limnic effect are site-specific it is not possible to compare the absolute isotope values from different sites possibly influenced by sea spray effect and/or limnic influence.

The minimum sea spray effect in $\delta^{13}C_{carb}$ and $\delta^{18}O_{carb}$ values of animal bones was approximated as 32.8% and 62.8%, respectively. These percentage values are markedly higher than found for the minimum sea spray effect (16.5%) in $\delta^{34}S_{coll}$ (Göhring et al., 2015). This can be explained by several reasons:

Both carbon and oxygen atoms have a much lower atomic weight compared to sulphur atoms. For carbon in marine aerosols three essential forms are known, namely carbonate-bicarbonate, organic carbon, and elemental carbon (Duce, 1983). Marine oxygen is transported to terrestrial environments as dissolved CO_2, HCO_3^-, and CO_3^{2-}. Marine SO_4^{2-} in sea spray, on the other hand, affects terrestrial sulphur isotope ratios. Since the molecular weight of CO_2, HCO_3^-, and CO_3^{2-} is lower than that of SO_4^{2-}, it can be assumed that even more carbon and oxygen compounds are transported inland by sea spray aerosols. This would explain the large impact of the sea spray on carbonate carbon and oxygen stable isotopic values. Furthermore, each component contains much more oxygen atoms than both carbon and sulphur atoms. Thus, the higher impact of sea spray on $\delta^{18}O_{carb}$ can probably be explained by the higher probability of a marine oxygen being transported to the terrestrial environment.

Sea spray aerosol causes an input of marine carbon and oxygen into terrestrial ecosystems. Marine carbon and oxygen can be incorporated into plants by uptake of both CO_2 and HCO_3^- via plant roots. Tracer studies showed that carbon of bicarbonate taken up by plant roots is transported to the shoot and seeds and used for sugar and starch synthesis (e. g. Bedri et al., 1960; Overstreet et al., 1940; Schäfer, 1988; Zamanian et al., 2017). Seawater carbonate has a $\delta^{18}O$ value of about 0‰ (by definition) and marine bicarbonate exhibits a carbon isotope value of about 1‰ (Mook, 2000), thus introducing quite high isotopic ratios into the terrestrial environment. Bicarbonate can also be transformed into CO_2 (and vice versa) by plant carbonic anhydrases (e. g. Badger and Price, 1994; Tiwari et al., 2005), therefore playing a major role in photosynthetic reactions. Furthermore, bicarbonate serves for anaplerotic replenishment reactions in the citric acid cycle by producing e. g. oxaloacetate and malate (Melzer and O'Leary, 1987). Consequently, the marine signal caused by sea spray is incorporated into the carbon and oxygen isotopic signal of plants.

Carbon and oxygen isotopes in bone apatite are derived from blood bicarbonate and serum CO_2. generated by the energy metabolism (Ambrose et al., 1997; Gannes et al., 1998; Sullivan and Krueger, 1981). Consequently, sea spray can influence the isotopic signal in consumers' tissues due to digestion of carbohydrates produced in plants influenced by sea spray (see above). These carbohydrates will have a distinct marine isotopic signal in their carbon and oxygen stable isotopes caused by the sea spray effect. $\delta^{18}O$ values mainly reflect the oxygen input into body water, namely drinking water, metabolic oxygen (from food), and atmospheric oxygen (Bryant and Froelich, 1995; Kohn, 1996). However, since $\delta^{18}O_{carb}$ is less affected by drinking water (compared to $\delta^{18}O$ of bone phosphate), but rather influenced by the oxygen isotopic composition of the ingested food (water) (Kirsanow and Tuross, 2011), incorporation of these plants will cause a marine signal accumulating in the tissues, and therefore even bone apatite, of the consumers. Nevertheless, even uptake of drinking water only slightly influenced by the sea spray can result in more positive oxygen isotope values. Furthermore, this marine signal will not only be present in herbivores, but also carnivores and omnivores feeding on both the herbivorous animals and influenced plants, respectively. Consequently, this would explain a possible way of incorporation of the sea spray effect in bone (and even tooth) apatite of mammals.

However, carbonate oxygen and even carbon stable isotopic values can also be influenced by several other effects. Environmental parameters such as temperature, water source, and geographical location

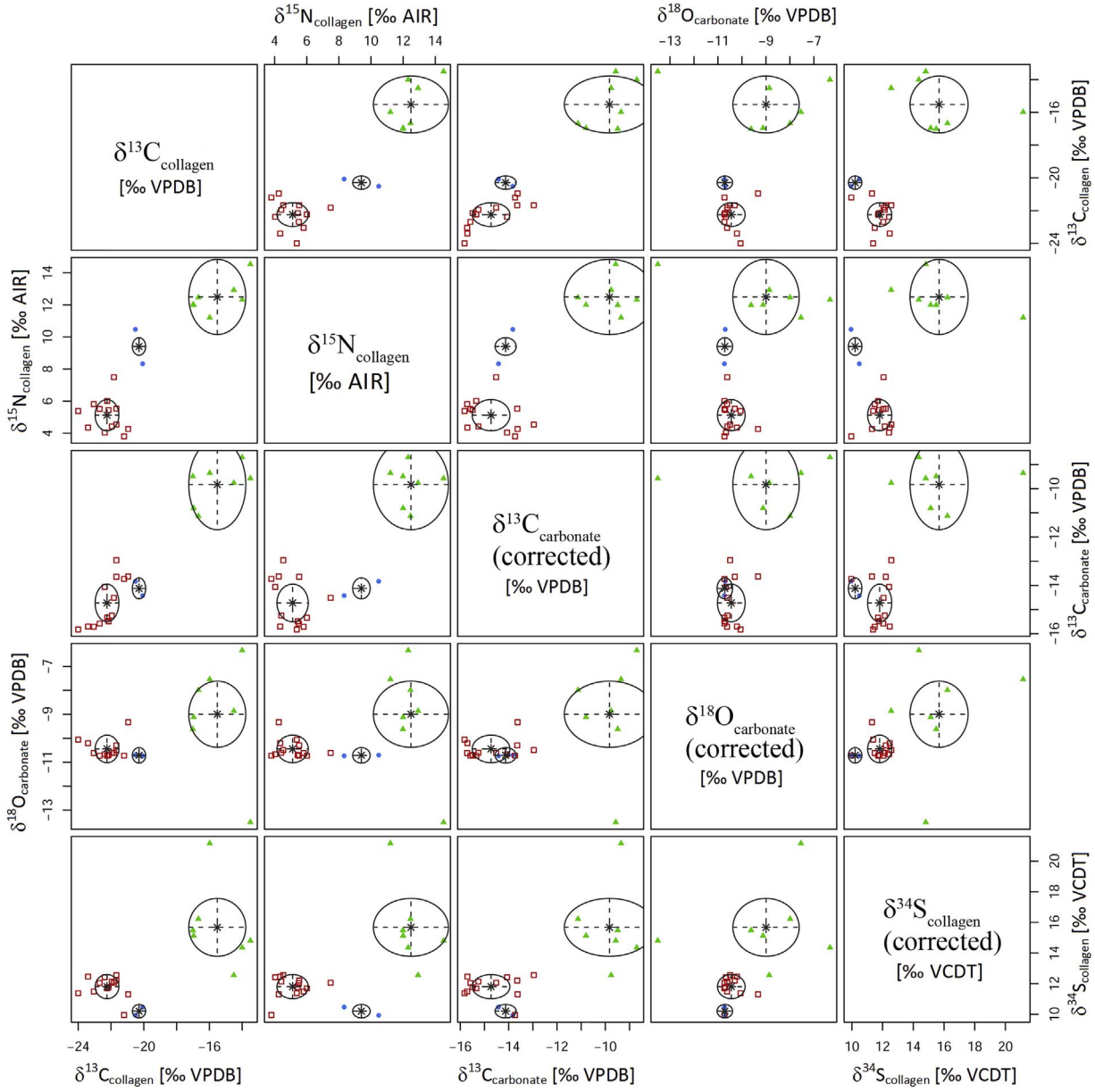


Fig. 5. Pairwise bivariate plot of the resulting three clusters performed with the "VEI" model (Fraley et al., 2012) on the five-dimensional data set (terrestrial mammals corrected for sea spray in $\delta^{13}C_{carb}$ (32.8%) and $\delta^{18}O_{carb}$ (62.8%), and $\delta^{34}S_{coll}$ (16.5%)); cluster 1: dots, cluster 2: squares, cluster 3: triangles.

Table 7
GMM clustering results performed with three clusters and the "VEI" model according to the five-dimensional data set corrected for the sea spray effect.

Cluster 1	Cluster 2	Cluster 3
Aurochs Hb47	Cattle Hb57	Common seal Hb25
Roe deer S11	Hare Hb2	Common seal KHb30
	Hare Hb3	Common seal KHb31
	Hare S24	Grey seal S17
	Hare S28	Saddleback seal Hb30
	Horse Hb92	Saddleback seal KHb25
	Red deer S3	Walrus KS76
	Red deer S4	
	Red deer S5	
	Roe deer Hb40	
	Roe deer Hb41	
	Sheep Hb74	
	Sheep Hb79	

Terrestrial mammals were corrected for sea spray in $\delta^{13}C_{carb}$ (32.8%), $\delta^{18}O_{carb}$ (62.8%), and $\delta^{34}S_{coll}$ (16.5%).

Table 8
Mean and standard deviation (sd) of the corrected five–dimensional data set according to GMM clustering performed with three clusters.

	Cluster 1	Cluster 2	Cluster 3
Mean			
$\delta^{13}C_{coll}$ [‰ VPDB]	− 20.29	− 22.24	− 15.52
$\delta^{15}N_{coll}$ [‰ AIR]	+ 9.41	+ 5.12	+ 12.49
$\delta^{13}C_{carb}$ [‰ VPDB]	− 14.13	− 14.73	− 9.83
$\delta^{18}O_{carb}$ [‰ VPDB]	− 10.71	− 10.44	− 9.00
$\delta^{34}S_{coll}$ [‰ VCDT]	+ 10.21	+ 11.81	+ 15.67
sd			
$\delta^{13}C_{coll}$ [‰ VPDB]	0.41	0.71	1.67
$\delta^{15}N_{coll}$ [‰ AIR]	0.55	0.95	2.26
$\delta^{13}C_{carb}$ [‰ VPDB]	0.44	0.79	1.87
$\delta^{18}O_{carb}$ [‰ VPDB]	0.32	0.57	1.36
$\delta^{34}S_{coll}$ [‰ VCDT]	0.44	0.83	1.96

Terrestrial mammals were corrected for sea spray in $\delta^{13}C_{carb}$ (32.8%), $\delta^{18}O_{carb}$ (62.8%), and $\delta^{34}S_{coll}$ (16.5%).

will have an influence on the isotopic signature of the examined animals. This needs to be emphasised since we cannot exclude that some of the individuals recovered at Haithabu and Schleswig were imported from other environments. Nevertheless, a quantifiable part of the overlap of the isotopic signature of herbivorous and marine mammals was probably caused by sea spray.

Interestingly, domesticated individuals seem to be more affected by sea spray compared to wild mammals. Most domesticated mammals were grouped into the first ("mixed") cluster, while the majority of wild ones was grouped into the second cluster (Table 1, Fig. 2). Domesticated individuals were kept on open fields. On the other hand, wild animals, such as deer, spend most of their life in the forest or on forest glades. The forest canopy will probably keep off a major part of the sea spray before reaching the ground and thus the food and the drinking water of herbivores. Consequently, sea spray can have a higher impact on domesticated individuals who feed on the grass and drink water in an open field, maybe close to the coast, directly affected by sea spray. As there were probably several different pastures around Haithabu and Schleswig, the different individuals are not all influenced in the same way but show a varying impact in their isotopic signature. This can be explained by pastures with different distance from the coast, a different water and food source, and partly maybe even import.

Even after correction for the minimum sea spray effect, several terrestrial mammals still exhibit a "too marine" carbon, oxygen, and sulphur stable isotopic signature, respectively. This can partly be explained by a marine diet, e.g. fish scraps in the case of red fox S18/S21 and pig Hb65, or a marine habitat (common otter 32 FO1Hepi). Herbivores (hare S28, sheep Hb76/Hb77/Hb78/Hb79) should not exhibit such a diet-related high marine influence. Although seaweed feeding is documented for sheep from Orkney (Scotland), causing an enrichment with ^{18}O (Balasse et al., 2009), there is no evidence for those practices in Haithabu and Schleswig. However, we cannot exclude that these individuals were not local but rather imported to the trade centres. In particular, trade with game had increased in the early time period of Schleswig (Heinrich, 1991). This could have been the case for hare S28 and would maybe explain the enrichment in ^{18}O with an (uncorrected) oxygen stable isotopic value of $-6.27‰$. A closer look at the sex distribution of some taxa in the archaeofaunal assemblage reveals a high amount of males and a high proportion of very young individuals among the sheep faunal remains, leading Hüster (1990) to assume a deliberate import of sheep. The enrichment in ^{18}O detected in sheep could however also be explained by the drinking water source of these individuals. If they had drunk brackish water this would have led to an enrichment in ^{18}O (Doppler et al., 2009).

After correction for the minimum sea spray effect in both the carbonate and sulphur isotopes, several animals remained in the "mixed" cluster 1 (Tables 5 and 7). This could be explained in different ways:

Some of these animals exhibited elevated $\delta^{15}N$ values compared to the remaining animals of our sample, especially rabbit S30 ($\delta^{15}N = +10.98‰$), aurochs Hb47 ($\delta^{15}N = +10.48‰$), fallow deer S6 ($\delta^{15}N = +9.43‰$) and roe deer S11 ($\delta^{15}N = +8.33‰$). This could be caused by a limnic influence probably triggered by aquatic insects (see above), representing an ecological vector for material from aquatic to terrestrial ecosystems (Gratton et al., 2008), resulting in an enrichment in ^{15}N in terrestrial vertebrates. This influence was approximated as at least 20.8% in the $\delta^{15}N_{coll}$ value of aurochs Hb47 (Göhring et al., 2015). The real effect might be even larger because in our present data set, we detected an even higher $\delta^{15}N_{coll}$ value in rabbit S30 (Table S2). However, as it is the case for the sea spray approximation procedure, an approximation of the local limnic influence should not be performed on domesticated animals anyway, because of a lack of information regarding details of the herding management. In the European Middle Ages, rabbits were frequently imported as live animals and kept in "rabbit gardens" (Heinrich, 1991), where they were deliberately fed by their owners.

The limnic influence not only affected the nitrogen isotopic values of the highlighted mammals of cluster 1 (Tables 5 and 7), but possibly the entire terrestrial animals of our data set. Therefore, it is necessary to correct the $\delta^{15}N_{coll}$ values of all terrestrial animals accordingly (Göhring et al., 2015).

The importance of the simultaneous correction for both sea spray and limnic influence was demonstrated by forcing the data into two clusters only, following the assumption that all individuals were perfectly corrected and afterwards showed their original and unaffected isotopic signature. With correction for the minimum sea spray effect in the carbonate fraction only, two herbivores (aurochs Hb47, rabbit S30) were still clustered together with marine mammals, probably because of their extreme $\delta^{15}N_{coll}$ values (Fig. S3). Additional correction for the limnic influence (20.8%; Göhring et al., 2015) solved this problem and led to a pure terrestrial and a pure marine cluster, respectively (Fig. S4).

In the case of the examined marine mammals, also two outliers that grouped into cluster 1 (Table 1) were detected: common seal Hb23 and sperm whale Hb50. As already explained above, this might be due to either a freshwater impact or an origin from more northern (colder) regions (Heinrich, 1991; Marshall et al., 2007; Mook, 1971; Reichstein, 1974, 1991).

Due to the choice of the end members for the calculation of the marine impact (m) some animals showed a marine impact higher than 1 or lower than 0 (Fig. S2). This can be especially critical for individuals with a negative impact value as we would desire a very small correction there because of a very low influence of the sea spray effect. However, the modulus in Eq. (4) causes a too high correction of the respective isotope value as already mentioned before. If the isotope values of these individuals are corrected for the sea spray effect, this would result in rather too low values compared to the original (uncorrected) values. However, if this is kept in mind when analysing the corrected isotope data this should not cause any problems. It might also be possible to exclude all animals which do not fall within the herbivorous range, however, this could have an influence on the isotope set-up of the whole data set. Hence, we recommend to keep and correct these certain isotope values but to be aware of a possibly too low corrected isotope signature in the case of those individuals. Including those animals in e.g. food web calculations, however, should be avoided in order to exclude wrong calculation results.

Discriminant analysis performed to validate the calculated GMM models resulted in low training and test errors in the case of the four-dimensional data set before and after correction for the sea spray indicative of an efficient model. These results are equivalent to both a low classification error in the trained data set (training error) and a low proportion of incorrectly classified individuals in the data set not used to train the model (test error) (see Section 2.2). Additional correction for the limnic influence on nitrogen stable isotopes, under the assumption of only two clusters, leads to even better training ($0.02\% \pm 0.17\%$) and test ($0.80\% \pm 2.07\%$) errors (see above). This result is superior to the one obtained on a GMM model with two clusters and corrected carbonate isotope values only (training error: $0.52\% \pm 0.90\%$; test error: $2.57\% \pm 2.84\%$), and also much better than the mathematically expected value of 50%. The five-dimensional data set showed training errors of 0.0% for both uncorrected and corrected data what is maybe an artificial result as we used labels resulting from the same GMM clustering procedure performed on the training set itself in the absence of known (semantic) labels (see above), what is however usually the case for isotope analyses. Nevertheless, meaningful clustering would result in a mean training error close to 0.0% anyway. The relatively high mean test errors of 21.4% (± 16.2%, uncorrected) and 17.1% (± 13.2%, corrected) calculated for the five-dimensional subsample are probably a result of the small data set of only 22 individuals. However, in both cases the results are better than expected (~33.3%). The even better training and test results after correction for the sea spray effect and the limnic influence, respectively, compared to the models using uncorrected isotope values further indicate the validity of the data correction procedure.

As mentioned before, the approximated sea spray effect of 38.2%, 68.2%, and 16.5% in $\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$, and $\delta^{34}S_{coll}$, respectively, is site-specific. Variability between different sites can be caused by e. g. variations in the distance from the coast, the salinity level of the sea water (brackish vs. marine), the prevailing wind direction, as well as the wind force at the sites under study. Nevertheless, the approximation method used in this study can be adopted for other data sets. It is also important to mention that our approximation method can also be performed on a much smaller data set. There are only few parameters prerequisite for an approximation of the local sea spray effect: At least one marine (piscivorous) mammal is needed, which has to be (probably) local to the study site. Furthermore, at least two terrestrial herbivorous wild mammals are needed to calculate the "marine impact" on terrestrial animals in the data set. A higher number of herbivores in the data set would naturally increase the power of the "marine impact". In order to approximate the local sea spray signal at least one herbivore has to exhibit an isotopic value higher than the lowest value found in the set of marine mammals. Consequently, a data set consisting of only at least three animal remains could potentially be enough to approximate the local minimum sea spray effect. It hast to be kept in mind that sea spray can be only approximated if an overlap of terrestrial herbivores and marine mammals is present. A missing overlap can, however, not be seen as evidence for an absent sea spray effect at the study site (see above) due to the phenomenon of "missing data".

5. Conclusions

We conclude that the sea spray effect is not only present in bone collagen $\delta^{34}S$, but also in carbon and oxygen stable isotopes of the bone carbonate. The presence of the effect in $\delta^{13}C_{coll}$ and $\delta^{15}N_{coll}$ could not be verified, however not firmly excluded. In the bone carbonate fraction, the sea spray effect amounts at least to 38.2% ($\delta^{13}C_{carb}$) and 68.2% ($\delta^{18}O_{carb}$), respectively. These values even exceed the approximated minimum sea spray effect influencing $\delta^{34}S_{coll}$ (16.5%) calculated previously (Göhring et al., 2015). We are positive that our approximation method can be adapted to other data sets after an adjustment to the site-specific background data.

While stable isotope analysis of archaeological animal bone finds is a useful tool for the reconstruction of palaeobiodiversity and changing environmental conditions over time, it is indispensable that the measurement data correctly reflect a taxon's habitat and feeding preferences. This is not given or granted in a complex palaeoecosystem like the one studied by us, where four basic ecological niches do exist (terrestrial, freshwater, estuarine, marine). Some species are more flexible in their habitat preference and e. g. tolerate different levels of salinity.

Both the detected sea spray effect in $\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$, and $\delta^{34}S_{coll}$, and the limnic influence on $\delta^{15}N_{coll}$ values are capable to obscure the isotopic signature of terrestrial taxa. These results are of importance for future studies on both prehistoric and recent materials in coastal environments. All stable isotopic signatures of terrestrial and freshwater animals living or feeding in coastal areas and of all terrestrial animals living in the vicinity of lakes are potentially affected by the sea spray effect and the limnic influence, respectively.

The GMM clustering performed on archaeological animal bone finds from a complex palaeoecosystem resulted in a clear separation of individual habitat and feeding preference that was impossible by univariate or bivariate statistics.

Since isotopic fractionation and mixing within an ecosystem generates compartments with characteristic isotopic signatures, isotopic labels shared by soil, water, plants, and animals are successfully used for the generation of isotopic landscapes. While in modern ecology stable isotopic signatures are used as labels which permit the reconstruction of the element flux through an ecosystem, palaeobiodiversity studies are dependent on the evaluation of such processes by hindsight. Since GMM clustering can be successfully applied to past environmental conditions, we suggest that such mathematical procedures will also be of benefit for modern ecology.

Acknowledgements

The stable isotope analysis was funded by the German Science Foundation, DFG (grant number Gr 959/15-1,2).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.palaeo.2017.11.057.

References

Ambrose, S.H., Butler, B.M., Hanson, D.B., Hunter-Anderson, R.L., Krueger, H.W., 1997. Stable isotopic analysis of human diet in the Marianas Archipelago, western Pacific. Am. J. Phys. Anthropol. 104, 343–361.
Badger, M.R., Price, G.D., 1994. The role of carbonic anhydrase in photosynthesis. Annu. Rev. Plant Physiol. Plant Mol. Biol. 45, 369–392.
Balasse, M., Mainland, I., Richards, M.P., 2009. Stable isotope evidence for seasonal consumption of marine seaweed by modern and archaeological sheep in the Orkney archipelago (Scotland). Environ. Archaeol. 14, 1–14.
Becker, C., Grupe, G., 2012. Archaeometry meets archaeozoology: Viking Haithabu and medieval Schleswig reconsidered. Archaeol. Anthropol. Sci. 4, 241–262.
Bedri, A.A., Wallace, A., Rhoads, W.A., 1960. Assimilation of bicarbonate by roots of different plant species. Soil Sci. 89, 257–263.
Bryant, J.D., Froelich, P.N., 1995. A model of oxygen isotope fractionation in body water of large mammals. Geochim. Cosmochim. Acta 59, 4523–4537.
Clementz, M.T., Koch, P.L., 2001. Differentiating aquatic mammal habitat and foraging ecology with stable isotopes in tooth enamel. Oecologia 129, 461–472.
Clementz, M.T., Goswami, A., Gingerich, P.D., Koch, P.L., 2006. Isotopic records from early whales and sea cows: contrasting patterns of ecological transition. J. Vertebr. Paleontol. 26, 355–370.
Clementz, M.T., Fox-Dobbs, K., Wheatley, P.V., Koch, P.L., Doak, D.F., 2009. Revisiting old bones: coupled carbon isotope analysis of bioapatite and collagen as an ecological and palaeoecological tool. Geol. J. 44, 605–620.
Craig, H., Gordon, L.I., 1965. Deuterium and oxygen 18 variations in the ocean and the marine atmosphere. In: Tongiorgi, E. (Ed.), Conference on Stable Isotopes in Oceanographic Studies and Paleotemperatures, Spoleto, pp. 9–130.
DeNiro, M.J., 1985. Postmortem preservation and alteration of in vivo bone collagen isotope ratios in relation to palaeodietary reconstruction. Nature 317, 806–809.
Doppler, S., Vohberger, M., Grupe, G., Peters, J., von Carnap-Bornheim, C., 2009. Siedlungskontinuität, Wirschaftswandel und Paläoökologie: Vorläufige Ergebnisse von Isotopenanalysen an Tierknochenfunden aus dem wikingerzeitlichen Haithabu und dem mittelalterlichen Schleswig. Beiträge zur Archäozoologie und Prähistorischen Anthropologie 7, 151–164.
Doppler, S., Vohberger, M., Von Carnap-Bornheim, C., Peters, J., Grupe, G., 2010. Biodiversity of archaeological fauna in the estuarine palaeoecosystem of the Schlei fjord, northern Germany: isotopic evidence. Documenta Archaeobiologiae 8, 21–70.
Duce, R.A., 1983. Biogeochemical cycles and the air/sea exchange of aerosols. In: Bolin, B., Cook, R.B. (Eds.), SCOPE 21 - the Major Biogeochemical Cycles and Their Interactions. John Wiley & Sons, Chichester, pp. 427–456.
Emeis, K.-C., Struck, U., Blanz, T., Kohly, A., Voß, M., 2003. Salinity changes in the central Baltic Sea (NW Europe) over the last 10000 years. The Holocene 13, 411–421.
Fraley, C., Raftery, A., Murphy, T.B., Scrucca, L., 2012. mclust version 4 for R: normal mixture modeling for model-based clustering, classification, and density estimation. Tech. Rep. 597, 1–57.
Fry, B., Sherr, E.B., 1984. $\delta^{13}C$ measurements as indicators of carbon flow in marine and freshwater ecosystems. Contrib. Mar. Sci. 27, 13–47.
Gannes, L.Z., Martinez del Rio, C., Koch, P., 1998. Natural abundance variations in stable isotopes and their potential uses in animal physiological ecology. Comp. Biochem. Physiol. A Mol. Integr. Physiol. 119, 725–737.
Göhring, A., Vohberger, M., Nehlich, O., Grupe, G., 2015. Approximation of the sea spray effect and limnic influence on $\delta^{34}S$ and $\delta^{15}N$ values of archaeological human and terrestrial and freshwater animal skeletal finds. Documenta Archaeobiologiae 12, 169–188.
Göhring, A., Mauder, M., Kröger, P., Grupe, G., 2016. Using Gaussian mixture model clustering for multi-isotope analysis of archaeological fish bones for palaeobiodiversity studies. Rapid Commun. Mass Spectrom. 30, 1349–1360.
Gratton, C., Donaldson, J., Zanden, M.J., 2008. Ecosystem linkages between lakes and the surrounding terrestrial landscape in Northeast Iceland. Ecosystems 11, 764–774.
Grupe, G., Heinrich, D., Peters, J., 2009. A brackish water aquatic foodweb: trophic levels and salinity gradients in the Schlei fjord, northern Germany, in Viking and medieval times. J. Archaeol. Sci. 36, 2125–2144.
Grupe, G., von Carnap-Bornheim, C., Becker, C., 2013. Rise and fall of a medieval trade centre: economic change from Viking Haithabu to medieval Schleswig revealed by stable isotope analysis. Eur. J. Archaeol. 16, 137–166.
Harbeck, M., Grupe, G., 2009. Experimental chemical degradation compared to natural diagenetic alteration of collagen: implications for collagen quality indicators for stable isotope analysis. Archaeol. Anthropol. Sci. 1, 43–57.

Heinrich, D., 1991. Untersuchungen an Skelettresten wildlebender Säugetiere aus dem mittelalterlichen Schleswig. Karl Wachholtz Verlag, Neumünster.
Hilberg, V., 2008. Hedeby: an outline of its research history. In: Brink, S. (Ed.), The Viking World. Routledge, London, New York, pp. 101–111.
Hobson, K.A., 1999. Tracing origins and migration of wildlife using stable isotopes: a review. Oecologia 120, 314–326.
Hüster, H., 1990. Untersuchungen an Skelettresten von Rindern, Schafen, Ziegen und Schweinen aus dem mittelalterlichen Schleswig. In: Ausgrabung Schild 1971–1975. Neumünster, Karl Wachholtz Verlag.
Kahle, D., Wickham, H., 2013. ggmap: spatial visualization with ggplot2. R J. 5, 144–161.
Kirsanow, K., Tuross, N., 2011. Oxygen and hydrogen isotopes in rodent tissues: impact of diet, water and ontogeny. Palaeogeogr. Palaeoclimatol. Palaeoecol. 310, 9–16.
Koch, P.L., 2007. Isotopic study of the biology of modern and fossil vertebrates. In: Michener, R., Lajtha, K. (Eds.), Stable Isotopes in Ecology and Environmental Science. Blackwell Publishing Ltd, pp. 99–154.
Kohavi, R., 1995. A study of cross-validation and bootstrap for accuracy estimation and model selection. In: Proceedings of the 14th International Joint Conference on Artificial intelligence. 2. pp. 1137–1143.
Kohn, M.J., 1996. Predicting animal $\delta^{18}O$: accounting for diet and physiological adaptation. Geochim. Cosmochim. Acta 60, 4811–4829.
Marshall, J.D., Brooks, J.R., Lajtha, K., 2007. Sources of variation in the stable isotopic composition of plants. In: Michener, R., Lajtha, K. (Eds.), Stable Isotopes in Ecology and Environmental Science, 2 ed. Blackwell Publishing, Malden, Oxford, Victoria, pp. 22–60.
Melzer, E., O'Leary, M.H., 1987. Anapleurotic CO_2 fixation by phosphoenolpyruvate carboxylase in C_3 plants. Plant Physiol. 84, 58–60.
Miyake, Y., Wada, E., 1967. The abundance ratio of $^{15}N/^{14}N$ in marine environments. Rec. Oceanogr. Work Jpn. 9, 37–53.
Mizota, C., Sasaki, A., 1996. Sulfur isotope composition of soils and fertilizers: differences between Northern and Southern hemispheres. Geoderma 71, 77–93.
Monaghan, J.M., Scrimgeour, C.M., Stein, W.M., Zhao, F.J., Evans, E.J., 1999. Sulphur accumulation and redistribution in wheat (*Triticum aestivum*): a study using stable sulphur isotope ratios as a tracer system. Plant Cell Environ. 22, 831–839.
Mook, W.G., 1971. Paleotemperatures and chlorinities from stable carbon and oxygen isotopes in shell carbonate. Palaeogeogr. Palaeoclimatol. Palaeoecol. 9, 245–263.
Mook, W.G., 2000. Environmental isotopes in the hydrological cycle: principles and application. In: Volume 1: Introduction. International Atomic Energy Agency and United Nations Educational, Scientific and Cultural Organization, Paris, Vienna.
Nehlich, O., 2015. The application of sulphur isotope analyses in archaeological research: a review. Earth-Sci. Rev. 142, 1–17.
Nehlich, O., Richards, M., 2009. Establishing collagen quality criteria for sulphur isotope analysis of archaeological bone collagen. Archaeol. Anthropol. Sci. 1, 59–75.
Overstreet, R., Ruben, S., Broyer, T.C., 1940. The absorption of bicarbonate ion by barley plants as indicated by studies with radioactive carbon. Proc. Natl. Acad. Sci. U. S. A. 26, 688–695.
Peterson, B.J., Howarth, R.W., 1987. Sulfur, carbon, and nitrogen isotopes used to trace organic matter flow in the salt-marsh estuaries of Sapelo Island, Georgia. Limnol. Oceanogr. 32, 1195–1213.
R Core Team, 2016. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.
Rees, C.E., Jenkins, W.J., Monster, J., 1978. The sulphur isotopic composition of ocean water sulphate. Geochim. Cosmochim. Acta 42, 377–381.
Reichstein, H., 1974. Ergebnisse und Probleme von Untersuchungen an Wildtieren aus Haithabu (Ausgrabung 1963–1964). In: Schietzel, K. (Ed.), Berichte über die Ausgrabungen in Haithabu: Bericht 7. Karl Wachholtz Verlag, Neumünster, pp. 103–144.
Reichstein, H., 1991. Die wildlebenden Säugetiere von Haithabu (Ausgrabungen 1966–1969 und 1979–1980). Karl Wachholtz Verlag, Neumünster.
Richards, M.P., Hedges, R.E.M., 1999. Stable isotope evidence for similarities in the types of marine foods used by late Mesolithic humans at sites along the Atlantic Coast of Europe. J. Archaeol. Sci. 26, 717–722.
Richards, M.P., Fuller, B.T., Molleson, T.I., 2006. Stable isotope palaeodietary study of humans and fauna from the multi-period (Iron Age, Viking and Late Medieval) site of Newark Bay, Orkney. J. Archaeol. Sci. 33, 122–131.
Schäfer, W., 1988. Pflanzenwachstum durch CO_2/HCO_3-Eintrag über die Wurzel. J. Agron. Crop Sci. 160, 228–234.
Schaffner, F.C., Swart, P.K., 1991. Influence off diet and environmental water on the carbon and oxygen isotopic signatures of seabird eggshell carbonate. Bull. Mar. Sci. 48, 23–38.
Schmidt, H.L., Robins, R.J., Werner, R.A., 2015. Multi-factorial in vivo stable isotope fractionation: causes, correlations, consequences and applications. Isot. Environ. Health Stud. 51, 155–199.
Schoeninger, M.J., DeNiro, M.J., 1984. Nitrogen and carbon isotopic composition of bone collagen from marine and terrestrial mammals. Geochim. Cosmochim. Acta 48, 625–639.
von Steinsdorff, K., Grupe, G., 2006. Reconstruction of an aquatic food web: Viking Haithabu vs. medieval Schleswig. Anthropol. Anz. 64, 283–295.
Sullivan, C.H., Krueger, H.W., 1981. Carbon isotope analysis of separate chemical phases in modern and fossil bone. Nature 292, 333–335.
Tanz, N., Schmidt, H.-L., 2010. $\delta^{34}S$-value measurements in food origin assignments and sulfur isotope fractionations in plants and animals. J. Agric. Food Chem. 58, 3139–3146.
Tcherkez, G., Tea, I., 2013. $^{32}S/^{34}S$ isotope fractionation in plant sulphur metabolism. New Phytol. 200, 44–53.
Tiwari, A., Kumar, P., Singh, S., Ansari, S.A., 2005. Carbonic anhydrase in relation to higher plants. Photosynthetica 43, 1–11.
Virginia, R.A., Delwiche, C.C., 1982. Natural ^{15}N abundance of presumed N_2-fixing and non-N_2-fixing plants from selected ecosystems. Oecologia 54, 317–325.
Wadleigh, M.A., Schwarcz, H.P., Kramer, J.R., 1994. Sulphur isotope tests of seasalt correction factors in precipitation: Nova Scotia, Canada. Water Air Soil Pollut. 77, 1–16.
Wakshal, E., Nielsen, H., 1982. Variations of $\delta^{34}S$ (SO4), $\delta^{18}O$ (H2O) and Cl/SO_4 ratio in rainwater over northern Israel, from the Mediterranean Coast to Jordan Rift Valley and Golan Heights. Earth Planet. Sci. Lett. 61, 272–282.
Wickham, H., 2009. ggplot2: Elegant Graphics for Data Analysis. Springer, New York.
Zamanian, K., Pustovoytov, K., Kuzyakov, Y., 2017. Carbon sources in fruit carbonate of *Buglossoides arvensis* and consequences for 14C dating. Radiocarbon 59, 141–150.
Zazzo, A., Monahan, F.J., Moloney, A.P., Green, S., Schmidt, O., 2011. Sulphur isotopes in animal hair track distance to sea. Rapid Commun. Mass Spectrom. 25, 2371–2378.